

Ali Parchekani

Toronto, ON, Canada • (647) 766-7304 • aliparchekan@gmail.com

[LinkedIn](#) • [GitHub](#) • [Google Scholar](#)

Academic Teaching Profile

PhD Candidate in Electrical and Computer Engineering at the University of Toronto with extensive teaching experience across mathematics, statistics, computer science, programming, physics, artificial intelligence, machine learning, data science, networks, and engineering fundamentals. Supported instruction across 40+ distinct University of Toronto courses (100+ TA/course-staff appointments) and independently designed and taught pre-university enrichment courses in statistics, critical thinking, AI, and game design. Teaching consistently draws strong, specific evaluations praising clarity, step-by-step explanation, and individualized support. Teaching style combines intuition-first explanation, structured problem solving, activity-based learning, and exam-focused support.

Education

Ph.D. in Electrical and Computer Engineering — Communications

University of Toronto, Toronto, ON | January 2021 – Expected August 2026

Thesis: Sensing and Scanning with Reconfigurable Intelligent Surfaces (in progress). Candidacy achieved: Summer 2023.

Relevant coursework: Algorithms for Collective Decision Making (A), Combinatorial Methods (A+), Non-Linear Optimization (A), Detection and Estimation Theory (A+), Game Theory and Evolutionary Games (A).

M.Sc. in Electrical and Computer Engineering — Communication Networks

University of Tehran, Tehran, Iran | September 2018 – December 2020

Thesis: Neural-Network-Based Network Traffic Classification and Aggregate User-Behavior Prediction Using Inverse Reinforcement Learning.

GPA: 19.57/20 — ranked 1st in the department.

B.Sc. in Electrical Engineering — Telecommunications

University of Tehran, Tehran, Iran | September 2014 – September 2018

GPA: 18.78/20 (3.98/4) — ranked 3rd in the department.

Awards, Scholarships, and Funding

- **Doctoral Completion Award**, University of Toronto — awarded for two years in recognition of strong research progress toward thesis completion.
- **University of Toronto Doctoral Fellowship / PhD Funding Package** — competitive graduate funding awarded to support doctoral study.
- **University of Tehran Undergraduate Scholarship** — merit scholarship awarded for three consecutive years for top academic standing.

Teaching Effectiveness

Course evaluations and end-of-term student feedback consistently highlight clarity of explanation, individualized support, and a calm, encouraging classroom. Selected, anonymized evidence:

Game Studies / Games, Strategy, and AI (*instructor of record; pre-university, 16 students aged 15–18*)

- On the overall course rating, all responding students rated the course “**Excellent**” or “**Very good**,” and **none** rated it Fair or Poor.
- Every responding student would recommend the course to another student (“Yes” or “Maybe”; no “No” responses).
- “*Explained the concepts clearly and helped students during activities.*”

- “You always help and guide us through the game-developing process.”
- “If he doesn’t know the answer, he will research it and tell us.”
- “Provided helpful, timely feedback on our project.”
- “I felt I learned more from asking you questions than from following the projector screen — I’m glad I had you as the instructor for this class.”
- Constructive feedback (acted on mid-course): students asked for more live coding demonstrations and clearer walk-throughs of the GDevelop tooling before examples — prompting a shift toward structured studio sprints in later sessions.

MAT A29 – Calculus I for the Life Sciences *(Teaching Assistant, tutorials & quizzes, Winter 2025)*

- “My TA went over specific homework questions that relate to the quizzes; his way of explaining is straightforward and helpful.”
- “The steps were broken down and explained thoroughly — there was reasoning behind each step.”
- “Thoroughly explained the concepts each tutorial and held office hours to clear any doubts before quizzes.”
- “Always available for questions and frequently ensured students knew it; prepared with reading-guide questions to go over.”
- “Very understanding and kind, with great communication skills; explains things that are easily comprehensible.”
- Constructive feedback: a student noted pacing could occasionally feel rushed — a useful prompt toward deliberately slower, signposted explanations.

Instructor Experience

ISA Instructor — Game Studies / Games, Strategy, and AI (PRU037H1S)

University of Toronto, Pre-University Programs / New College | April–May 2026

- Designed and independently taught a 15-session, 45-hour Game Studies and AI course to 16 students aged 15–18, combining foundations of games and strategy, introductory AI concepts, and hands-on game development in GDevelop.
- Built the full course — lesson flow, activities, technical demonstrations, and project milestones — culminating in playable, student-built game prototypes with a visible AI capability (rule-based NPC, pattern-tracking opponent, or adaptive difficulty).
- Used game design as an accessible bridge into AI, strategy, logic, debugging, computational thinking, and creative technical problem solving.

ISA Instructor — Critical Thinking in Statistics

University of Toronto, Pre-University Programs / New College | July–August 2024 (two offerings)

- Taught two offerings guiding pre-university learners through statistical reasoning, data interpretation, probability/statistics foundations, hypothesis testing, introductory machine-learning concepts, and evidence-based conclusions.
- Designed activity-based lessons around real-world questions, peer-consensus reasoning, data collection, and clear written conclusions.
- Created structured instructional materials: session plans, slides, worksheets, assignments, project guides, and guided-practice resources.

ISA Instructor — Applied Mathematics in Statistics; Science Across the Spectrum

University of Toronto, Pre-University Programs / New College | June–July 2025

- Supported pre-university students in quantitative reasoning, mathematical foundations for statistics, and structured problem solving; connected mathematical intuition to applied statistical thinking.
- Supported interdisciplinary STEM learning through structured lessons, worked examples, and guided academic support.

University Teaching Assistantship and Course-Staff Experience

Supported instruction across 40+ distinct University of Toronto courses (100+ TA/course-staff appointments) at UTM, UTSC, and St. George, spanning computer science, programming, networks, AI/ML, statistics, probability, mathematics, physics, electrical and computer engineering, and data science. Repeatedly reappointed to high-demand quantitative courses, including Bayesian statistics, stochastic processes, computer networks, combinatorics, and introductory computer science.

Computer Science, Programming, and Software: CSC108, CSC148, CSCA08, CSCA20, CSCA48, CSC110, CSCB63, CSCC63, ECE244.

Computer and Communication Networks: CSC358, CSC457, ECE361, ECE1541.

AI, Machine Learning, and Data Science: CSC311, CSC384, CSCD84, STA315, MIE223.

Statistics, Probability, Bayesian Statistics, Stochastic Processes, Time Series: STA107, STA256, STA260, STA313, STA315, STA348, STA360, STA413, STA457, STA2202.

Mathematics: MAT102, MAT132, MAT135, MAT223, MAT224, MAT236, MAT406, MATA29, MATA33, MATA36, MATB44, MATC44, ECE1657.

Physics, Signals, and Engineering Fundamentals: PHYA10, PHYA22, PHYB57, PHY131, PHY132, PHY180, ECE216.

Teaching Approach and Instructional Design

- Teaches from experience toward abstraction: starts from a concrete problem, dataset, system, or game, names the underlying concept, then applies it to new situations.
- Breaks difficult topics into intuition, formal methods, worked examples, and independent practice; uses activity-based learning and exam-focused preparation as course goals require.
- Builds structured notes, worksheets, examples, project guides, and review materials that connect homework, quizzes, projects, and exams.
- Designs assessment to reward explanation, application, and revision — not only final answers — using rubrics, project checkpoints, and feedback that makes the reasoning visible.
- Treats teaching as iterative design: adjusts pacing and structure in response to where students slow down, while protecting the intended learning outcome.

Private Tutoring and Academic Mentorship

- **Private STEM Tutor — Calculus and Physics.** Tutored pre-university students individually through in-person one-on-one sessions, using diagnostic assessment, step-by-step concept repair, guided problem solving, and targeted practice.
- **Research Mentorship.** Mentored undergraduate students during the master's research period, supporting technical development, research problem solving, and quantitative/computational workflows.

Selected Research and Publications

PhD researcher in wireless sensing, radar signal processing, reconfigurable intelligent surfaces, optimization, and applied machine learning, with peer-reviewed IEEE publications and industry-facing research deliverables.

Peer-reviewed conference papers

1. A. Parchekani and S. Valaee, "Reconfigurable Intelligent Surface Assisted Sensing and Localization using the Swendsen-Wang and Evolutionary Algorithms," IEEE VTC 2023-Fall.
2. A. Parchekani and S. Valaee, "Sensing and Localization using Reconfigurable Intelligent Surfaces and the Swendsen-Wang Algorithm," IEEE ICC Workshops 2022.
3. A. Parchekani and S. Valaee, "Sensing via Orthogonal Time Frequency Space Signalling and Reconfigurable Intelligent Surface," IEEE PIMRC 2022.
4. A. Parchekani, M. Johnny, and S. Valaee, "Model-Based Deep Learning Tuning of Reconfigurable Intelligent Surface for OFDM Radar Interference Mitigation," IEEE PIMRC 2025.

Preprints

- A. Parchekani, S. Nouri, V. Shah-Mansouri, and S. P. Shariatpanahi, "Classification of Traffic using Neural Networks by Rejecting: A Novel Approach in Classifying VPN Traffic," arXiv:2001.03665.
- A. Parchekani and M. Johnny, "Reconfigurable Intelligent Surface for OFDM Radar Interference Mitigation," arXiv:2502.14730.

Teaching Technologies, Training, and Technical Skills

Teaching tools: Quercus/Canvas, Piazza, Gradescope, Jupyter notebooks, GDevelop, Python, MATLAB, LaTeX/Overleaf, Zoom, PowerPoint, Google Classroom, GitHub, autograders, simulation demos.

Professional training: UofT TA training; TATP Policies & Procedures; TATP Labs & Practicals; Managing Distressing or Complex Student Situations; Graduate Engineering Student Community Resources; lab safety training.

Programming & scientific computing: Python, SQL, TypeScript, JavaScript, C, C++, MATLAB, LaTeX; NumPy, SciPy, pandas, Matplotlib, PyTorch, scikit-learn, TensorFlow.

Research methods: Optimization, MCMC, evolutionary algorithms, statistical inference, experiment design, ablation studies, signal processing, wireless sensing.